Can Large Language Models Understand Context?

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Motivation - Importance of Context Understanding

- Understanding context is key to understand human language, and an essential ability for Large Language Models (LLMs)

- Aspects of discourse understanding
  - Resolving the relations between words/phrases/discourse units
  - Identifying carry-over information
  - Recognizing discourse-specific phenomena such as ellipsis

Motivation - Context Understanding

- Plenty of benchmarks and datasets address various facets of language understanding

- None of these benchmarks cater to tasks that demand a nuanced comprehension of linguistic features within a provided context

Motivation – LLM Challenges in Practice

- The size of LLMs hinders the deployment of large models to personal devices and restricts the on-device performance of language understanding tasks

- It is not yet clear whether large, compressed models out- or under-perform their smaller counterparts when understanding context
- Previous work has not delved into a linguistic analysis to identify where the model efficacy wanes.

The Context Understanding Benchmark

Coreference Resolution

Nominal & eventual reference

WSC 273
OntoNotes

Doc
In the summer of 2005, a picture that people have long been looking forward to started emerging with frequency in various major **Hong Kong** media. With **their** unique charm, **these well-known cartoon images** once again caused **Hong Kong** to be a focus of worldwide attention ...

Clusters
{Hong Kong, Hong Kong}
{their, these well - known cartoon images}

Implicit Discourse Relation Classification

Relations between discourse units

PDTB-3

Arg1
An ancient stone church stands amid the fields,x

Arg2
the sound of bells cascading from its tower, calling the faithful to evensong.xx

Relation Label
Temporal

Dialogue State Tracking

Entity carryover within context

MultiWoz

user: [TEXT]
bot: [TEXT]

⋮

user: I am looking for a train that leaves on **Thursday** going to **Cambridge**.

⋮

bot: [TEXT]
user: [TEXT]

train-day: Thursday
train-destination: Cambridge

Query Rewriting

Ellipsis and reference

MuDoCo
QReCC
InCar
GECOR
CANARD

User: Try to reach Forbes now.
Bot: Forbes at Washington Post? Or Forbes of Publishing Division?
User: Publishing Division.

Rewrite
Forbes of Publishing Division.

Experiments Settings

- Three model families
  - OPT (Zhang et al., 2022)
  - LLaMA (Touvron et al., 2023)
  - GPT (OpenAI, 2023)
- Various model sizes
  - OPT: 125M, 350M, 1.3B, 2.7B, 6.7B, 13B, 30B
  - LLaMA: 7B, 13B, 30B
  - GPT: 3.5-turbo

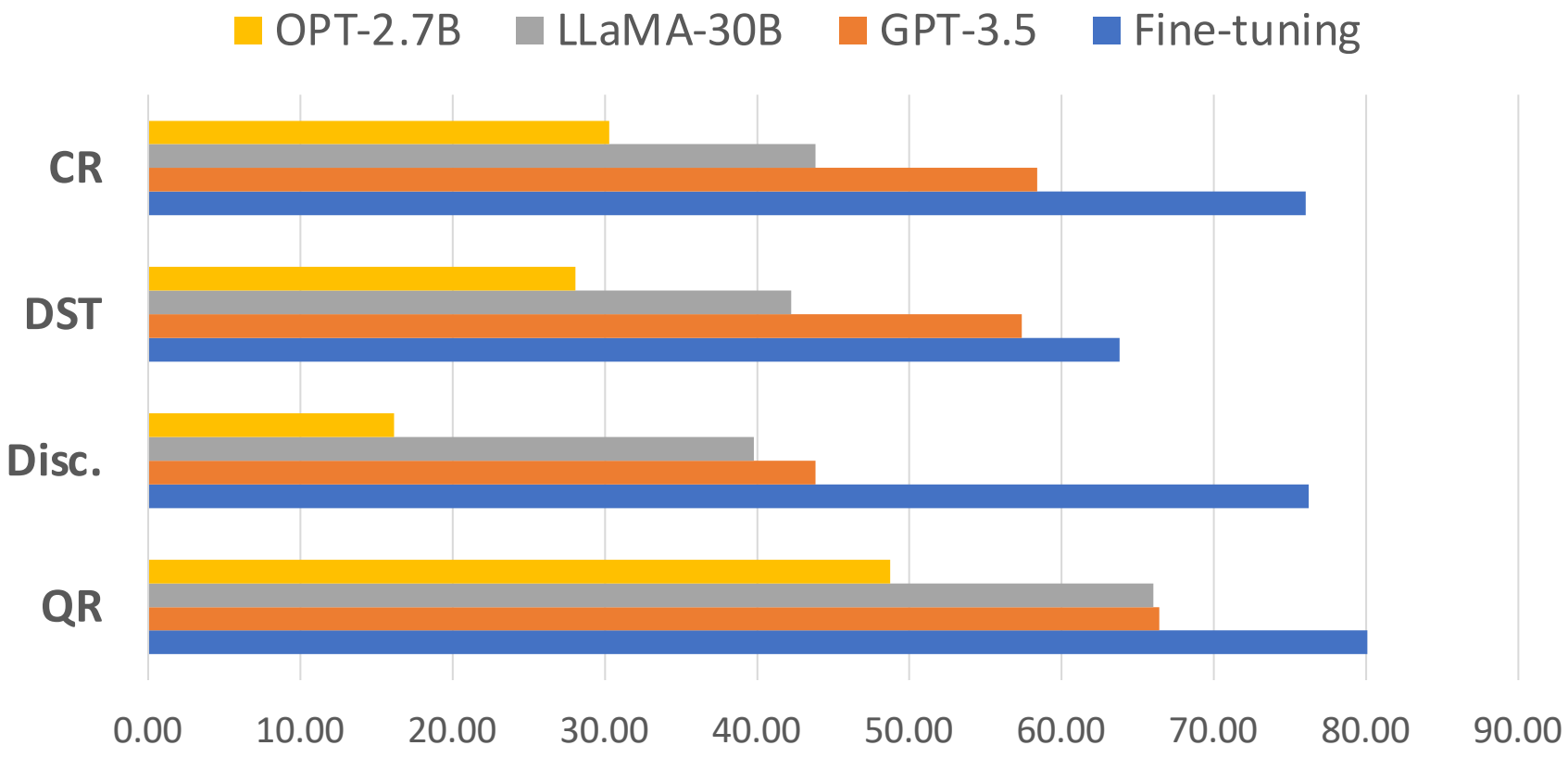
Experiments Settings

- In-context learning (ICL) (Brown et al., 2020a)
- Classification
  - Coreference resolution
  - Implicit discourse relation classification
- Generation
  - Dialogue state tracking
  - Query rewriting

Instruction: Given two arguments and a list of connective words, please select the most likely connective between two arguments.
[Relation Description]
Input:
Arg 1: Amcore, also a bank holding company, has assets of $1.06 billion.
Arg 2: Central's assets are $240 million.
Question: What is the connective that best describes the relation between two arguments?
Choices:
A. Temporal B. Contingency C. Comparison D. Expansion
Answer: *C*

Table 3: A PDTB example of prompt and *answer*.

Experiments



Experiments

Task	Dataset	Metrics	OPT				LLaMA			GPT	FT
			125M	350M	1.3B	2.7B	7B	13B	30B	3.5-turbo	
CR	WSC273	Acc	58.24	66.67	76.19	77.66	86.81	89.38	89.01	88.64	N/A
	OntoNotes	MUC	12.66	7.58	13.21	8.29	10.31	31.80	33.56	56.32	77.26
		B^3	53.80	52.26	53.54	52.41	52.20	58.43	58.66	68.20	73.43
		$CEAF_{\phi 4}$	31.09	29.49	31.40	30.10	32.63	38.00	39.27	50.72	74.46
		Avg. F1	32.52	29.78	32.72	30.27	31.71	42.74	43.83	58.41	76.03
DST	MultiWOZ	JGA	11.11	27.96	26.61	28.08	32.30	28.12	42.24	57.40	63.79
Disc.	PDTB-3	Acc	10.04	10.04	10.04	16.15	17.16	26.01	39.77	43.83	76.23
QR	MuDoCo	BLEU	0.46	0.36	7.02	49.20	41.12	61.15	66.51	57.14	80.31
		ROUGE	1.52	12.18	10.98	65.61	56.07	74.78	77.88	79.37	92.01
	QReCC	BLEU	4.53	31.27	26.35	40.09	28.19	38.64	58.68	55.24	58.67
		ROUGE	13.91	58.18	53.10	68.32	48.27	56.40	78.74	79.98	81.75
	InCar	BLEU	0.00	7.66	12.71	27.42	28.20	42.13	48.58	63.66	88.45
		ROUGE	3.41	28.76	30.45	49.63	49.96	56.73	64.18	83.51	95.24
	GECOR	BLEU	0.20	26.40	26.32	49.99	53.27	66.30	73.80	63.34	82.56
		ROUGE	4.06	42.13	42.57	65.89	69.23	80.99	86.03	79.00	92.63
	CANARD	BLEU	2.61	19.39	24.24	34.66	21.34	29.32	47.24	47.12	57.46
		ROUGE	9.82	45.63	49.36	62.73	38.17	46.61	69.73	74.61	81.06

Table 5: Few-shot results of two open-sourced models and GPT-3.5 on the context understanding benchmark. The results with the best number of few-shot examples are reported for each task. Fine-tuning (FT) results serves as a reference when evaluating LLMs' capability under ICL setup.

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WSC is easy for LLMs - expected, given it's a binary classification task

OntoNotes is more difficult
- All models struggle, perform far away from the fine-tuned (FT) model
- Error analysis: model tends to predict singleton clusters

Experiments

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Only GPT catches up with FT model

Error analysis:
- LLMs over trigger slots or predict wrong value

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All LLMs perform significantly worse than the fine-tuned (FT) model

Error analysis:
- All LLMs tend to predict the same relation class for every example

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Smaller model cannot even generate reasonable output

Error Analysis:
- Generate next sentence instead of rewriting
- Rewrite the wrong user query
- Simply copy instead of rewriting

Experiments Settings - Quantization

- 3-bit post-training quantization
  - GPTQ (Frantar et al., 2022)
- Models
  - LLaMA: 7B, 13B, 30B

Dense vs. Quantized

- Quantization leads to fluctuated drops in performance across the four tasks

Dataset	Metrics	7B-D	30B-Q	30B-D
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Table 6: Comparison between dense and quantized models. Dense LLaMA-7B and 3-bit quantized LLaMA-30B share similar memory and disk requirements. **D** represents dense model and **Q** denotes quantized model.

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  - Marginal performance drop (~1.7 points) in *WSC273, MultiWOZ, and CANARD*

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Dense vs. Quantized

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  - Marginal performance drop (~1.7 points) in WSC273, MultiWOZ, and CANARD
  - Significant performance drops in *rest datasets*

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Query Rewriting – A Case Study

- Two comparisons
  - Between model families
  - Between dense and quantized models

Query Rewriting – OPT vs. LLaMA

- Prior works (Beeching et al., 2023)
  - LLaMA consistently outperforms OPT under the same model size

Query Rewriting – OPT vs. LLaMA

- Prior works (Beeching et al., 2023)
  - LLaMA consistently outperforms OPT under the same model size
- Our findings (query rewriting)
  - **Model size ~7B**
    - **OPT > LLaMA**

Dataset	6.7/7B		13B		30B	
	O.	L.	O.	L.	O.	L.
Mudoco	53.1	41.1	55.2	61.1	55.2	66.5
	71.8	56.0	72.1	74.7	71.5	77.8
QReCC	46.6	28.1	43.7	38.6	43.8	58.6
	73.4	48.2	71.6	56.4	71.9	78.7
InCar	40.3	28.2	41.9	42.1	44.6	48.5
	64.8	49.9	62.6	56.7	65.3	64.1
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Table 7: Comparison between OPT (O.) and LLaMA (L.) across five query rewrite datasets. For each dataset, the first and second rows represent BLEU and ROUGE scores respectively.

Query Rewriting – OPT vs. LLaMA

- Prior works (Beeching et al., 2023)
  - LLaMA consistently outperforms OPT under the same model size
- Our findings (query rewriting)
  - Model size ~7B
    - OPT > LLaMA
  - **Model size ~13B**
    - **OPT ≈ LLaMA**

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- Our findings (query rewriting)
  - Model size ~7B
    - OPT > LLaMA
  - Model size ~13B
    - OPT ≈ LLaMA
  - **Model size ~30B**
    - **OPT < LLaMA**

Dataset	6.7/7B		13B		30B	
	O.	L.	O.	L.	O.	L.
Mudoco	53.1	41.1	55.2	61.1	55.2	66.5
	71.8	56.0	72.1	74.7	71.5	77.8
QReCC	46.6	28.1	43.7	38.6	43.8	58.6
	73.4	48.2	71.6	56.4	71.9	78.7
InCar	40.3	28.2	41.9	42.1	44.6	48.5
	64.8	49.9	62.6	56.7	65.3	64.1
GECOR	58.8	53.2	60.9	66.3	58.2	73.8
	75.7	69.2	78.3	80.9	76.1	86.0
CANARD	43.8	21.3	37.5	29.3	41.3	47.2
	72.0	38.1	66.0	46.6	69.3	69.7

Table 7: Comparison between OPT (O.) and LLaMA (L.) across five query rewrite datasets. For each dataset, the first and second rows represent BLEU and ROUGE scores respectively.

Query Rewriting – Dense vs. Quantized

- Two types of errors
  - Error type 1: repeat the last query
  - Error type 2: language modeling

- Example

`User`: what is the name of india pakistan border line

`Bot`: The Radcliffe Line was the boundary demarcation line between the Indian and Pakistani portions of the Punjab and Bengal provinces of British India.

`User`: who created the radcliffe line

`Bot`: The Radcliffe Line was named after its architect, Sir Cyril Radcliffe, who was the joint chairman of the two boundary commissions for the two provinces.

`User`: when was the line published

Gold answer: when was the radcliffe line published

Prediction 1 (repeat the last query): when was the line published

Prediction 2 (language modeling): 1947

Query Rewriting – Dense vs. Quantized

- Manual error annotations
- Memory and disk requirements
  - 7B Dense ≈ 30B Quantized

- Comparison
  - 7B Dense
  - 30B Dense
  - 30B Quantized

Query Rewriting – Dense vs. Quantized

- Repeat errors (lower the better)
  - 30B Dense < 7B Dense
  - 30B Quantized ≈ 7B Dense

- 3-bit post-training quantization reduces the model's ability to comprehend the context

Type	Dataset	7B D	30B Q	30B D
Repeat	MuDoCo	260	247	194
	QReCC	86	90	26
	InCar	17	15	8
	GECOR	59	62	37
	CANARD	47	44	32
	Total	469	458	297
LM	MuDoCo	71	29	16
	QReCC	80	28	16
	InCar	19	20	15
	GECOR	6	1	0
	CANARD	127	76	59
	Total	232	125	106

Table 9: Number of the major two types errors on three LLaMA models (7B dense, 30B quantized, and 30B dense) found in query rewriting. *Repeat* stands for repeat-the-last-query error and *LM* denotes language modeling error.

Query Rewriting – Dense vs. Quantized

- LM errors (lower the better)
  - 30B Dense < 7B Dense
  - 30B Quantized ≈ 30B Dense
  - < 7B Dense

- 3-bit post-training quantization maintains the ICL capability that allows models to rewrite the user query successfully rather than performing language modeling task

Type	Dataset	7B D	30B Q	30B D
Repeat	MuDoCo	260	247	194
	QReCC	86	90	26
	InCar	17	15	8
	GECOR	59	62	37
	CANARD	47	44	32
	Total	469	458	297
LM	MuDoCo	71	29	16
	QReCC	80	28	16
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Table 9: Number of the major two types errors on three LLaMA models (7B dense, 30B quantized, and 30B dense) found in query rewriting. *Repeat* stands for repeat-the-last-query error and *LM* denotes language modeling error.

Conclusion

- Introduce a contextual understanding benchmark designed to assess the performance of LLMs
- LLMs under ICL struggle with nuanced linguistic features within this challenging benchmark, exhibiting inconsistencies with other benchmarks
- 3-bit post-training quantization reduces the general understanding capacity of context across the 4 tasks